Vector Space

Let V be a nonempty set of certain objects, which may be vectors, matrices, functions or some other objects. Each object is an element of V and is called a **vector**.

Field: set of real or complex numbers (scalars)

~~Assume empty~~ Assume that two algebraic operations

(i) vector addition and (ii) scalar multiplication

are defined on elements of V.

If the vector addition is defined as the usual addition of vectors, then

$$a + b = (a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots + a_n + b_n)$$

If the scalar multiplication is defined as the usual scalar multiplication of a vector by the scalar α, then

$$\alpha a = \alpha(a_1, a_2, \dots, a_n) = (\alpha a_1, \alpha a_2, \dots, \alpha a_n).$$

Definition. A nonempty set V is called a vector space over the field F if the following conditions are satisfied

1. $a + b \in V$ for all a, b in V
2. $a + b = b + a$ for all a, b in V (commutative property)
3. $a + (b + c) = (a + b) + c$ for all a, b and c in V (associative property)
4. there exists an element 0 in V such that $a + 0 = a$ for all a in V
5. for each a in V there exists an element $-a$ in V such that $a + (-a) = 0$
6. αa is in V for all α in F and a in V
7. $\alpha(\beta a) = (\alpha\beta) a$ for all α, β in F and a in V
8. $\alpha(a + b) = \alpha a + \alpha b$
9. $(\alpha + \beta) a = \alpha a + \beta a$, $\alpha, \beta \in F$ [or α, β in F]
10. $1 . a = a$, 1 being the identity element in F.

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The properties defined in 1 and 6 are called the closure properties.
When these two properties are satisfied, we say that the vector space is closed under the vector addition and scalar multiplication.
The vector addition and scalar multiplication defined above need not always be the usual addition and multiplication operators. Thus, the vector space depends not only on the set V of vectors, but also on the definition of vector addition and scalar multiplication on V.

When the scalar field F is the set of real numbers $\mathbb{R}$, the vector space is called a real vector space. When the scalar field is the set of complex numbers $\mathbb{C}$, the vector space is called a complex vector space.

If even one of the above mentioned properties is not satisfied, then V is not a vector space.

Examples

i) Real vector space $\mathbb{R}^n$. Let V be the set of all ordered n-tuples $\{(a_1, a_2, \dots, a_n) : a_i \in \mathbb{R}\}$, with the following operations

$$(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1+b_1, a_2+b_2, \dots, a_n+b_n)$$

$$\alpha(a_1, a_2, \dots, a_n) = (\alpha a_1, \alpha a_2, \dots, \alpha a_n), \quad \alpha \in \mathbb{R}.$$

Then the conditions 1-10 are satisfied. Therefore V is a real vector space and it is denoted by $\mathbb{R}^n$.
Here $(0, 0, \dots, 0)$ is the zero element.

(ii) The set of all complex numbers $\{a+ib : a, b \in \mathbb{R}, i = \sqrt{-1}\}$ over the field $\mathbb{R}$.

Let + be a composition on $\mathbb{C}$, called 'addition' defined by

$$(a+ib) + (c+id) = (a+c) + i(b+d)$$

and the 'multiplication' operator is defined by

$$c(a+ib) = (ca) + i(cb), \quad c \in \mathbb{R}$$

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Then the conditions 1 - 10 are satisfied. Therefore the set of complex number $\mathbb{C}$ over the field $\mathbb{R}$ is a vector space.

ii) Let V be the set of all $m \times n$ matrices over $\mathbb{R}$.

Let + be a composition on V, called 'addition', defined by

$$(a_{ij})_{m\times n} + (b_{ij})_{m\times n} = (a_{ij} + b_{ij})_{m\times n}$$

and the 'multiplication of matrices by real numbers' be defined by

$$c\,(a_{ij})_{m\times n} = (c a_{ij})_{m\times n}, \quad c \in \mathbb{R}.$$

Then the conditions 1 - 10 are satisfied. Therefore V is a real vector space.

Ex. The set V of all polynomial degree n is not a vector space.

Consider two elements P_n & Q_n from V

$$P_n = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n$$

$$Q_n = -x^n + b_1 x^{n-1} + b_2 x^{n-2} + \dots + b_n$$

Now $P_n + Q_n = (a_1 + b_1)x^{n-1} + (a_2 + b_2)x^{n-2} + \dots + (a_n + b_n)$

$P_n + Q_n$ is a polynomial of degree $(n-1)$ and does not belong to V, so property 1 (closure property) does not hold. Thus V is not a vector space

Ex. Let V be the set of all ordered pairs (x, y), where x, y are real numbers. Let $a = (x_1, y_1)$ and $b = (x_2, y_2)$ be two elements in V. Define the addition as

$$a + b = (x_1, y_1) + (x_2, y_2) = (2x_1 - 3x_2,\ y_1 - y_2)$$

and the scalar multiplication as

$$\alpha(x_1, y_1) = (\alpha x_1/3,\ \alpha y_1/3).$$

Show that V is not a vector space.

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Solution

(i) $(x_1, y_1) + (x_2, y_2) = (2x_1 - 3x_2,\ y_1 - y_2)$

$(x_2, y_2) + (x_1, y_1) = (2x_2 - 3x_1,\ y_2 - y_1)$

$\therefore\ (x_1, y_1) + (x_2, y_2) \neq (x_2, y_2) + (x_1, y_1)$

Therefore, property 2 does not hold.

(ii) Let $a, b, c \in V$

$a = (x_1, y_1), \quad b = (x_2, y_2), \quad c = (x_3, y_3)$

$$a + (b+c) = (x_1, y_1) + \left((x_2, y_2) + (x_3, y_3)\right)$$
$$= (x_1, y_1) + \left(2x_2 - 3x_3,\ y_2 - y_3\right)$$
$$= \left(2x_1 - 6x_2 + 9x_3,\ y_1 - y_2 + y_3\right)$$

$$(a+b) + c = \left((x_1, y_1) + (x_2, y_2)\right) + (x_3, y_3)$$
$$= (2x_1 - 3x_2,\ y_1 - y_2) + (x_3, y_3)$$
$$= (4x_1 - 6x_2 - 3x_3,\ y_1 - y_2 - y_3)$$

$$a + (b+c) \neq (a+b) + c.$$

$\therefore$ Associative property does not hold.

(iii) $1 \cdot a = 1 \cdot (x_1, y_1) = (x_1/3,\ y_1/3) \neq (x_1, y_1) = a$

There does not exist any identity element

Hence, V is not a vector space. [Note - need to show any one of the condition]

Theorem. Let V be a vector space over a field F.

(i) For any scalar $k \in F$ and $0 \in V$, $k0 = 0$.

(ii) For $0 \in F$ and any vector $u \in V$, $0u = 0$.

(iii) If $ku = 0$, where $k \in F$ and $u \in V$, then $k = 0$ or $u = 0$.

(iv) For any $k \in F$ and $u \in V$, $(-k)u = k(-u) = -ku$.

Subspace: Let V be a vector space over a field F and let W be a nonempty subset of V. Then W is a subspace of V if W is itself a vector space over F with respect to the operations of vector addition and scalar multiplication on V.

The way in which one shows that any set $W \subseteq V$ is a vector space is to show that W satisfies the ~~eight~~ ten properties/axioms of a vector space.

Theorem. Suppose W is a subset of a vector space V. Then W is a subspace of V if the following two conditions hold:

(a) The zero vector 0 belongs to W.

(b) For every $u, v \in W$ and $\alpha \in F$ (i) The sum $u+v \in W$, (ii) $\alpha u \in W$.

or

(b') For every $u, v \in W$ and $\alpha, \beta \in F$, the linear combination $\alpha u + \beta v \in W$.

Example 1. Let V be a vector space over a field F. Then V itself is a subspace of V. This subspace is called the improper subspace of V.

The set consisting only of the null vector (zero vector) 0 of V forms a subspace of V. This subspace is called the trivial subspace of V.

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Example 2. Let S be the subset of $\mathbb{R}^3$ defined by $S = \{(x,y,z) \in \mathbb{R}^3 : y = z = 0\}$.

(i) $(0,0,0) \in S \Rightarrow S$ is nonempty and S contains zero vector

Let $u = (x_1, 0, 0)$, $v = (x_2, 0, 0) \in S$; $x_1, x_2 \in \mathbb{R}$

Let $\alpha, \beta \in \mathbb{R}$. Then $\alpha u + \beta v = \alpha(x_1, 0, 0) + \beta(x_2, 0, 0)$

$= (\alpha x_1 + \beta x_2, 0, 0) \in S$,

since $\alpha x_1 + \beta x_2 \in \mathbb{R}$.

This proves that S is subspace of $\mathbb{R}^3$.

Example 3. Let S be the subset of $\mathbb{R}^3$ defined by $S = \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}$

$(0,0,0) \in S$

Let $\alpha = (x_1, y_1, z_1)$, $\beta = (x_2, y_2, z_2) \in S$

Then $x_1^2 + y_1^2 = z_1^2$ and $x_2^2 + y_2^2 = z_2^2$

Now $\alpha + \beta = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$. $\alpha + \beta$ may not belong to S

because $(x_1 + x_2)^2 + (y_1 + y_2)^2$ may not be equal to $(z_1 + z_2)^2$.

For example, let $\alpha = (3, -4, 5)$, $\beta = (-3, 4, 5)$.

Then $\alpha, \beta \in S$ but $\alpha + \beta = (0, 0, 10) \notin S$

So, S is not a vector space.

Theorem. The intersection of two subspaces of a vector space V over a field F is a subspace of V.

Note # The intersection of a family of subspace of V is a subspace of V

The union of two subspaces of V is not, in general, a subspace of V.

For example, let us consider two subspaces S and T of the vector space $\mathbb{R}^3$ where $S = \{(x,y,z) \in \mathbb{R}^3 : y = 0, z = 0\}$, $T = \{(x,y,z) \in \mathbb{R}^3 : x = 0, z = 0\}$.

~~Let~~

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